

BHUJI KHOLA HYDROELECTRIC PROJECT



BHUJI KHOLA HYDROELECTRIC PROJECT (7.96 MW)

DHORPATAN MUNICIPALITY BAGLUNG

DESK STUDY REPORT

FOR

SURVEY LICENSE

Submitted By:

**TOPKEGOLA HYDROPOWER PVT. LTD.
MADHYAPURTHIMI -9, RADHERADHE, BHAKTAPUR, NEPAL**

JULY, 2025

SALIENT FEATURES OF THE PROJECT

1. General

Name of the Project	: Bhuji Khola Hydroelectric Project
Type of the Project	: Run-of-River (ROR) Hydropower Project
Name of River	: Bhuji Khola

2. Project Location

District/ Province	: Baglung, Gandaki Province
Project Location/MC	: Dhorpatan Municipality, Burtibang, Baglung
Nearest Market	: Burtibang Bazar, Mid-Hill Highway

3. Geographical Co-ordinates

Latitude	: 28° 21' 34" N - 28° 23' 01" N
Longitude	: 83° 06' 39" E - 83° 08' 22" E

4. Hydrology

Catchment Area	: 118.53 Km ²
Mean Annual Discharge	: 9.49 m ³ /s
Design Discharge (Q 40 %)	: 5.42 m ³ /s
Average Annual Precipitation	: 1964 mm

5. Power and Energy

Gross Head	: 180.0 m
Net Head at Full Flow	: 170.33 m
Installed Capacity	: 7.96 MW
Generated Energy per Annum	: 8.22 GWh (Dry) and 32.49 GWh (Wet)
Total: 40.71 GWh Energy (after Outage)	

6. Headworks Structures

Diversion Weir	: 44.0 m; Free Overflow, Concrete weir
Height of Weir	: 6.0 m from river bed
Weir Crest Elevation	: 1460.0 masl

Under-sluice	: 1 opening, 3.60 m (W) × 2.80 (H) m
Side Intake	: 2 openings, 2.0 (W) m × 1.6 (H) m
Approach Canal	: 25.0 (L) × 2.40 (W) × 1.50 (H) m; Rectangular Surface
Type of Settling Basin	: Surface Type with 2 Bays
Size of Each Bay	: 40.0 (L) m × 4.0 (B) m × 3.4 – 3.60 (D) m
Trapp Efficiency	: 90% of particle size greater than 0.2 mm

7. Headrace Pipe

Type	: Mild steel circular pipe
Pipe Length	: 2460 m
Internal Diameter	: 1.80 m Ø, Circular, (6 to 12 mm thickness)

8. Surge Tank

Type	: Surface, Concrete, with surge pipe
Size	: 4.0 m Dia. size,
Height of Surge	: 27.0 m

9. Steel Penstock Pipe

Type of Materials	: Mild Steel circular Pipe
Penstock Length	: 1310 m
Pipe Diameter (Internal)	: 1.75 to 1.70 m Ø, Circular, (10 mm to 28 mm thickness)

10. Power House

Type	: Surface
Size	: 21.0 (L) m × 9.0 (B) m × 9.0 (H) m
Tail Water Elevation	: 1280.0 masl
Length of Tailrace	: 50.0 m up to Bhuji Khola, Rectangular Box

11. Turbine

Turbine Type	: Horizontal Shaft Francis Turbine
Number of Generating Units	: 2
Capacity of each Unit	: 3.98 MW

Rated design discharge	: 2.71 m ³ /s for each unit
Efficiency	: 92.0%

12. Generators

Type	: Brushless synchronous- 3 phases, 50 Hz
Efficiency	: 97%
No of Unit	: 2
Rated Output per Unit	: 4.08 MVA

13. Transformer

Type	: 3 Phase, Oil Immersed, 33 kV (3 phase)
Nos. of Units	: 1 (One)
Rated Capacity per Unit	: 8.07 MVA
Power Factor	: 0.85
Efficiency	: 99.00%

14. Project Cost Estimate

Tentative Total Cost of the Project: NRs. 990.08 million Rs.

Payback Period	: 8.18 Years
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15. Transmission Line: The generated electricity will be evacuated 33 kV 3-phase transmission line of approximately 7.50 Km up to the Burtibang NEA Substation, Baglung.

16. Access Road: There is an existing black topped road through the project site; about 4.0 Km Road is required to be upgraded for project construction.

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1 INTRODUCTION

1.1 General

Nepal is a landlocked country in Southeast Asia surrounded by two big Asian nations - China in the north and India in east, south and west. Eight out of fourteen above eight-thousand-meter peaks in the world lies in the northern part of the country with flatter Terai plains in the south. Due to the fast-flowing rivers in the mountainous terrain with high gradients, Nepal has huge potential of hydropower development.

Nepal is suffering from energy crises and affecting in daily life of the people and the economic development of the country as a whole. Realizing the deficiency in power generation, the Government of Nepal intends to develop hydropower projects in an economical, efficient, and sustainable manner to meet the growing energy demands in the country as well to export.

Legislation in Nepal provides for the development and operation of hydropower projects by either private developers or NEA, under terms of survey and production license arrangements with Govt. of Nepal. Electricity Act, of 1992 and 2001 has facilitated wide business opportunities for local and foreign investors for developing hydropower projects. Domestic commercial banks have also shown their interest in investing in hydropower projects as a priority sector investment. This policy has encouraged private investors to put up hydropower projects and meet the growing demand for power in the country.

Nepal Electricity Authority has announced its policy to purchase the power produced by private developers. The price for the power generated from the small projects is fixed at NRs.4.80 for the wet months and NRs. 8.40 for the dry months. Poush to Chaitra (four months) are considered dry months. Solely the NEA manages the power industry in Nepal. Per capita electricity consumption of about 180 kWh/year is among Asia's lowest. But, the average electricity tariff of about Rs.8.5 /kWh is one of the highest in the region.

1.2 Project Background

The developer for this project is **Topkegola Hydropower Private Limited**, located in Madhyapur Thimi Municipality-09, Radheradhe, Bhaktapur, Nepal. They have entrusted **Hints Consult Pvt. Ltd.** with conducting the project study and preparing desk study reports. Relevant information about the project was collected and analyzed to gather the necessary data for this desk study. A report has been compiled to present the findings of the study at its initial stage. All activities undertaken to complete the study are outlined in the following chapters.

Bhuji Khola originates from the upper ranges of the western mountains and reaches Bayali Dhuri. This is the proposal for the survey license for the development of a ROR-type hydropower project in Bhuji Khola, Dhorpatan Municipality ward no. 7, Burtibang, Baglung district.

The project site is about 7.50 Km away from the nearest market, Burtibang, one of the major market center of Mid-hill highway, to Bhuji Khola, the headworks site.

1.3 Objectives of the Study

The main objective of the desk study is the collection and preliminary evaluation of available information and data on the topography, geology, hydrology, and accessibility of Bhuji Khola as well as the existing and tentative plans for general layout and transmission lines in the vicinity of the project area.

The desk study includes the general scale development of the project, selection of the general arrangement of structures, and determination of the general parameters of major structures and equipment.

The scope of work is as follows:

- To study available topographical maps and various literature on Geology, Hydrology, and Metrology related to the project area
- Preparation of the plan and concept of the project
- Study the various components of the project and prepare the project layout scheme
- Calculation of power and energy generation
- Project layout in Topo-map
- Prepare a basic scheme for the civil works, hydro-mechanical and electromechanical equipment, and transmission line
- Carry out tentative cost estimation of the project based on the present price for the project study for the feasibility study.
- To make necessary recommendations and conclusions about the project
- Preparation of Desk study report and further study.

1.4 Study Methodology

Different literature, maps, and reports are collected and studied. The important basic information on the topography of the proposed project sites was picked out from existing 1:25,000 scale topographical maps, GIS maps, and Google Earth. Other required data and

information were collected from the available Reports of the Budhiganga Nadi basin from the same area.

Besides technical requirements, it requires additional information from the Department of Electricity Development (DoED), the Ministry of Water Resources, Government of Nepal to get a survey and feasibility study license. So, the preparation of such documents also falls under the scope of this study. The additional documents regarding the feasibility study are:

- Scope of works
- Terms of reference
- Manpower
- Cost
- Schedule

All these documents are presented in the Appendix of this report. Relevant documents of the company are presented in the Appendix.

1.5 Available Information

For this study, topographical maps prepared by the Department of Survey and geological maps prepared by the Department of Mines and Geology were studied. Furthermore, more information was obtained from the Google Earth and Digital Elevation Model (DEM) data.

The Topo-sheets studied (available) are:

1. 2883 09D BURTIBAN (1:25 000)
2. 2883 09A JHIWAKHOLA (1:25 000)
3. GIS Maps
4. ARC GIS Earth
5. Google Earth Pro

2 HYDROLOGY

2.1 Physiographic Characteristics of Rivers

Bhuji Khola is a tributary of Badigad Khola in the Badigad River basin in Baglung. The basin has higher hills and mountainous catchments in northwestern Baglung. The catchment begins at the *Bayali Dhuri, Muthachaur Dada* with an elevation of 4043 masl. The total catchment area at the intake site of Bhuji Khola is 118.53 Km².

The proposed intake site is located at an elevation of about 1460.0 masl just below the confluence of Bhuji Khola and Gadi Khola and the powerhouse is at an elevation of 1280.0 masl just above the confluence of Bhuji Khola and Bhaisikharka Khola, about 3.5 Km downstream of the proposed intake site. Based on the topographical maps, there are no glaciers or lakes within the catchment area.

All of the above-mentioned drainage areas have been estimated based on the latest topographical maps compiled by the Department of Survey and Digital Elevation Model (DEM) data, GIS, and Arc GIS Earth, Google Earth Pro. The Google Earth Pro watershed map of Bhuji Khola in this project is presented in Figure 2-1.



Figure 2-1: Catchment Area (Google Earth Pro)

2.2 Hydrology and Stream Flow Data

2.2.1 Rainfall Data

Precipitation data is needed for the analysis of surface runoff and to know the nature of the catchment concerning the river flow. To further understand the effects of the elevation and location, the rainfall around the vicinity of the project area was also considered.

There is no availability of hydrological and hydro-meteorological data because DHM has no rainfall station at Bhuji Khola catchment. However, in the vicinity of Bhuji Khola catchment, DHM has four rainfall stations at Gajurkhani, Darbang, Musikot, and Bobang. These stations were selected for the hydrological study due to their proximity to Bhuji Khola catchments. Rainfall data are available for Gajurkhani, Darbang, and Bobang stations for the period between 1997 and 2000, and for Musikot station from 1971 to 2004. The summarized average annual precipitation is presented below Table 3-4.

Table 2-1: Average annual rainfall in the catchment, mm

Stations	Gurjakhani	Darbang	Musikot	Bobang	Areal Rainfall
Annual	1769.3	2243.2	2252.1	2655.7	1963.9
Monsoon Rainfall (Jun- Sep)	1505.6	1805.3	1766.8	2359.8	1534.2

This method gave annual rainfall as 1963.90 mm and annual monsoon rainfall as 1568.20 mm. Now, based on the above analysis it was conclude that the monsoon wetness index is related to monsoon rainfall as computed earlier. For further hydrological analysis, the monsoon wetness index was adopted as 2200 mm.

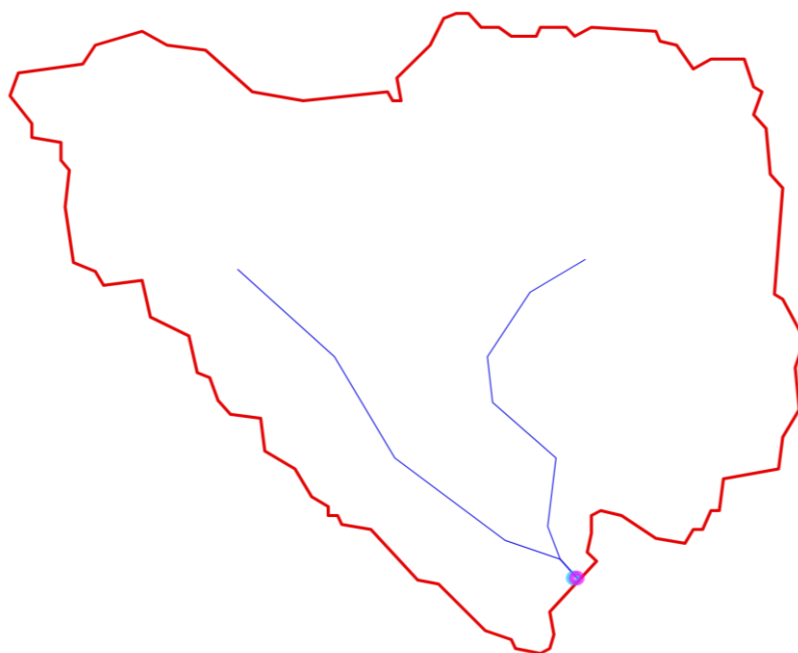


Figure 2-2 : Catchment Area Map in QGIS

2.2.2 Mean Monthly Flow

Stream flow records are important to estimate the flow. At the intake site, there are no records of flow measurements. Hence, for the discharge prediction, different hydrological analyses have been done and a reasonable monthly flow has been adopted.

1. MHSP Method

For the ungauged river station, the discharge prediction has been conducted using the Medium Hydropower Study Project (MHSP) methodology. This analysis leveraged data from several gauging stations to develop a comprehensive regional assessment, utilizing information spanning from 1974 to 2001. The gauging stations selected are recognized as reliable within the national context.

In this regional analysis, long-term average monthly flows were correlated with various basin characteristics. The regression analysis incorporated monthly flow data as well as maximum and minimum instantaneous flows collected from 66 hydrometric stations under the jurisdiction of the Department of Hydrology and Meteorology (DHM). The input variables employed were analogous to those utilized in the WECS/DHM methodology. This approach integrates both the monsoon wetness index and the average precipitation within the region in conjunction with the catchment area of the river.

2. Modified HYDEST Method

The river flow is predicted by the HYDEST method (later modified by WECS), which uses basin catchment, catchment characteristics, and monsoon index. From the average monthly flows obtained through regression, the average annual flow was developed for each of the ungauged sites. The monthly flow data were derived by considering a 2200 mm MWI, while the mean precipitation is 1964 mm. The monthly flow data obtained by the empirical formula are shown below.

3. Catchment Area Ratio Method

In this method, rivers having comparative physiographic characters with intake sites are chosen to generate the long series of discharge data and correlate with the catchment area.

Hence, the reference hydrology for the Bhuji Khola Project at the intake site has been derived from the gauging station of the Badigad River at Rudrabeni. DHM has a gauging station No. 417 located at Rudrabeni about 23 Km east-downstream from the proposed intake.

Badigad River gives the equivalent specific discharge compared to other rivers considered in the Kaligandaki Basin. It originates at an elevation of about 5000 m. The river flows in a southerly direction. A significant part of the Badigad Khola basin lies above 3500 m elevation.

The drainage area of Badigadh at Rudrabeni is 1940 Km². Mean monthly discharge data are available from DHM at Rudrabeni station is period of 1974-2001.

Looking at the physiographic conditions and proximity of the gauging stations, it would be more appropriate to use the discharge data from Rudrabeni station for deriving the streamflow at the intake site.

Therefore, the discharge available at Rudrabeni site can be used for a water availability study of Bhuji Khola intake site after transposing the data on the precipitation catchment area proportion basis. The formula used is given below:

$$Q_1 = \frac{P_1 A_1}{P_2 A_2} \times Q_2$$

Where,

P=Average Annual Precipitation (mm)

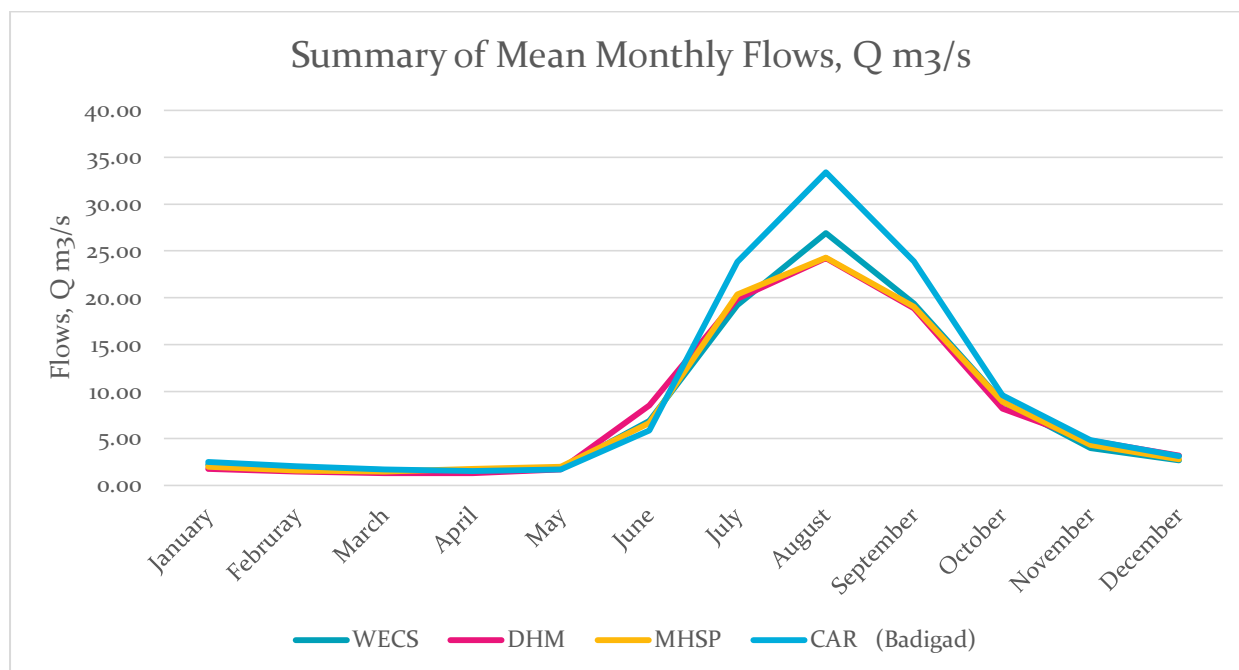
A=Basin Area (km²)

Q=River Discharge (m³/s).

P1 & A1 are for Gauging Station and P2 & A2 are for intake site

Table 2-2 : Summary of Mean Monthly Flows, Q m³/s

Month	WECS	DHM	MHSP	CAR (Badigad)
January	2.19	1.72	1.94	2.46
Februray	1.82	1.46	1.58	2.01
March	1.61	1.30	1.45	1.67
April	1.42	1.30	1.72	1.52
May	1.75	1.70	1.95	1.71
June	6.86	8.49	6.65	5.83
July	19.21	19.89	20.38	23.81
August	26.92	24.24	24.26	33.41
September	19.37	18.85	19.02	23.90
October	8.90	8.17	8.91	9.61
November	3.95	4.75	4.27	4.79
December	2.66	3.18	2.74	3.13
Average	8.06	7.92	7.91	9.49



2.3 Adopted Monthly Flow at the Intake Site

The above table shows that the derived long-term mean monthly flows at the intake site are comparable. The data from MHSP and Hydest are somewhat similar but are not comparable to direct flow measurements at the intake. This data is unreliable for long-term mean monthly flow when compared to the CAR method. For energy calculations, we need long-term hydrology data of the river. Therefore, it is not reliable to predict the long-term average flow of the intake using these methods.

The discharge obtained from the correlating catchment represents a more accurate result as basin characteristics and precipitation ratio are considered. For this reason, the adoption of the same catchment-correlated mean flow is most reliable. Therefore, we have adopted the CAR method by considering the precipitation of each gauging station and intake site in this study.

The average of the long-term mean monthly flow data, predicted by considering precipitation and the catchment area of both gauging stations, is used for the hydrological and energy calculations.

Table 2-3: Adopted Mean Monthly Flow for energy calculation (m³/s) at Intake Site

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	Average Flow
Flow (m ³ /sec)	2.46	2.01	1.67	1.52	1.71	5.83	23.81	33.41	23.90	9.61	4.79	3.13	9.49

2.4 Flow Duration Curve

A flow duration curve is a probability discharge curve that shows the percentage of time a particular flow is equaled or exceeded. The data, which correlates with 27 years of long-term flow data from the intake sites on the Badigadh River, is used to calculate flow percentiles. For the Bhuji Khola proposed intake site, this data has resulted in a flow of 5.42 m³/s at the 40th percentile. The flow duration curve at the intake site is shown in the figure below.

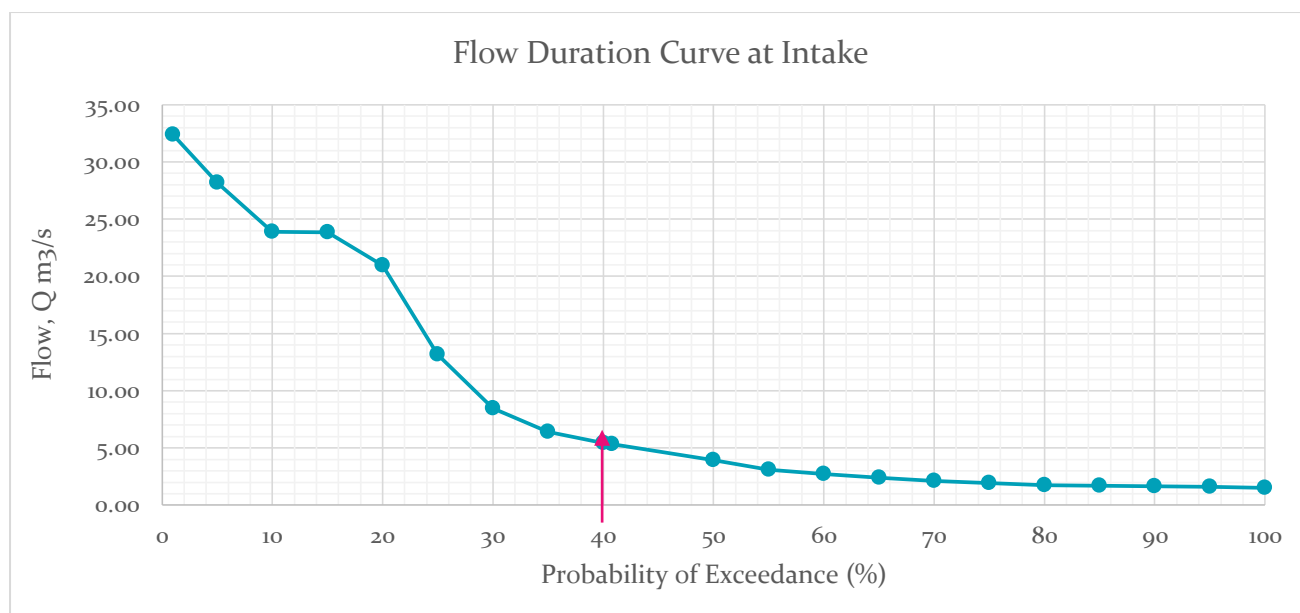


Figure 2-3: Flow Duration Curve

Table 2-4: Numerical Value of FDC at Intake Site

Probability of Exceedance (%)	Percentile Flow by CAR, Q m ³ /s	No. of days	Remarks
1	32.36	4	
5	28.18	19	
10	23.89	37	
15	23.84	55	
20	20.97	73	
25	13.16	92	
30	8.47	110	
35	6.40	128	
40	5.42	146	Q 40 %
41	5.33	149	
50	3.96	183	
55	3.10	201	
60	2.73	219	
65	2.39	238	
70	2.14	256	
75	1.93	274	
80	1.77	292	
90	1.67	329	
100	1.52	365	

3 DESCRIPTION OF THE PROJECT

3.1 Location

The project area lies in the Dhorpatan Municipality, the previous Budhiganga VDC of Baglung District. It is about 4.5 Km north-west of Burtibang, Baglung, the district headquarters. The Bhuji Khola Hydroelectric Project is envisaged as a run-of-river (ROR) scheme that utilizes the flow available at Bhuji Khola.

Geographically, the project area is located between the longitude of 83° 06' 39" E to 83° 08' 22" E and latitude of 28° 21' 34" N - 28° 23' 01" N. Physiographically, the project area belongs to the hilly to high mountain range and lies in between 1280.0 masl to 1460.0 masl of the Dhorpatan Municipality region.

The headworks are proposed below the confluence of Bhuji Khola and Gadi Khola. Similarly, the powerhouse is proposed at about 150m upstream of the confluence of Bhuji Khola and Bhaisikharka Khola.

3.2 General Description of the Project Components

The overall layout and the general arrangement of the main structure have been determined based on the available maps and other information. The configuration of the project mainly consists of a diversion weir, intake, approach canal, settling basin, headrace pipe, surge tank, penstock, powerhouse, and tailrace. All the structures are proposed at the left bank of the Bhuji Khola. The required structural features will be finalized after the feasibility study. The layout of the project (Topo-map sheet) is attached to this report.

3.2.1 Headworks

Diversion weir/Dam

A diversion dam/weir is proposed at an elevation of 1460.0 masl of Bhuji Khola. The weir shall be an ogee-shaped concrete free overflow structure to divert the required discharge for power generation into the intake and pass the flood flows downstream of the dam. The total length of the dam/weir will be about 44.0 m at the intake site. Whereas the dam height is proposed as 6.0m from the river bed. On the left bank of Bhuji Khola, sluiceways are proposed to flush the bed load. The size of the sluice openings is 3.60 m in width and 2.80 m in height. From the Dam, required discharge will be diverted to intake site with gravel trap arrangements.

A stilling basin shall be provided d/s of the weir and sluice to dissipate hydraulic energy during the high flows for the protection of the dam structure. Seepage protection measures shall be undertaken in the foundation of the dam.

Intake

Side intake is proposed on the left bank of the river of Bhuji Khola just after the weir. The intakes shall be provisioned as an orifice intake to divert the design discharge with 25 % flushing. Two intakes are proposed with an individual opening size of 2.0m width and 1.60 m height.

A gravel trap is proposed immediately after the intake with flushing facilities to settle the particle greater than 2 mm. The dimensions of the gravel trap will be 10.60m long, 3.0 width, and 2.20 m to 2.40 m tapering section of height.

Approach Canal

The approach canal transports water between the gravel trap and the settling basin. It also accommodates flood spillways. A fine trash rack will be placed at the beginning of the closed canal to check for trash.

An open gravity flow canal of 25.0 m long with a breadth of 2.40 m and a height of 1.50 m is required in the longitudinal slope 1 in 400 for Bhuji Khola.

Settling Basin

The settling basin is proposed to settle suspended sediments of particle size greater than 0.2 mm in diameter. Two bays are proposed so that the power plant can be operated partially during the flushing (with more than 90% efficiency) of one bay.

The preliminary calculations in this study proposed a dimension of 40.0 m in (L), 4.0 m (W) and vary depth from 3.4 to 3.60 m (D), including the transition length in the Bhuji Khola bank.

The sediment deposited at the bottom of the basin will be flushed out to the river through the flushing canal.

The size of the entire headworks structures will be finalized in the feasibility study. During the feasibility study, detailed financial and technical analysis will be carried out to select and optimize the entire headworks structures.

3.2.2 Waterways Alignment

Headrace Pipe

The waterway of design discharge from desander to surge; a low-pressure pipe of mild steel is proposed along the left bank of the Bhuji Khola, which seems shortest and favored in geological and topographical in this study.

The internal diameter of the pipe for the headrace and penstock has been determined through a preliminary optimization analysis that took into account head loss and cost.

A low-pressure headrace pipe with a circular-shaped section of 2460 m long of 1.80 m internal diameter is proposed for the surge from the Bhuji Khola settling basin. The total estimated headrace pipe will be supported by anchor blocks and piers. The length of pipe alignment might be varied after a detailed survey and measurement. The pipe is mild steel with 6 mm to 10 mm thickness.



Surge Tank

A surge tank is proposed to protect the horizontal penstock pipe from the effect of the water hammer caused by sudden changes in flow conditions and is proposed at the end of the headrace pipe.

The diameter of the tank is proposed at *Subjagau*. The dimension calculated in this study is 4.0 m and the height is estimated to be 27.0 m which shall be of pipe type of surge at Subjagau.

Penstock Pipe

A mild steel penstock pipe is proposed from the surge to convey the design discharge to the turbines in the powerhouse. The finished diameter of the pressure pipe is 1.75 to 1.70 m with 10 mm to 28 mm thickness. The estimated length of the pressure pipe is about 1280.0 m.

The alignment length diameter and side shall be further optimized in the feasibility study.

Anchor Blocks and Saddle Supports

Anchor blocks are required at the bends to support the pipe and saddle support piers will be provided along the straight sections of exposed penstock between anchor blocks to avoid

overstressing in the pipe. The required no. of anchor blocks and saddle supports are estimated during the feasibility study.

3.2.3 Powerhouse and Switchyard

A surface power house is proposed about 150m upstream of the confluence of Bhaisikharka Khola and Bhuji Khola at the left bank of Bhuji Khola. The powerhouse shall be of RCC structural building and steel truss and roofing material. The dimensions will be 21.0 m in length, 9.0 m in breadth, and 9.0 m in height.

The powerhouse is proposed to have two units of Horizontal Shaft Francis Turbines and generator units. The tailwater level of turbine elevation is proposed at 1280.0 masl. The efficiency of turbines will be 92.0 %. Each unit will have a 3.98 MW capacity of 2.71 m³/s design discharge.

The tailrace is about 50.0 m long up to the Bhuji Khola. A rectangular box type of tailrace canal with 2.40 m width and 1.25 m height will be required.

There will be 50 Hz, Brushless synchronous- 3-phase type 2 Generator units. The rated output per unit of the Generator is assumed as 4.08 MVA with a power factor of 0.85, assumed efficiency of generator units is 97%.

Power transformers are used to step up the generated power to evacuate the generated power to the nearest substation. The main transformers of the project shall have a double three-phase, outdoor, oil immersed, ONAN type of 8.07 MVA for stepping up the voltage with the efficiency of 99%.

3.2.4 Accessibility and Transmission

Accessibility

Access Road is considered an integral part of the development of hydropower projects. It plays a vital role not only for construction and maintenance purposes but also for the socio-economic development of the project area and the strengthening of the living standards of the people of the project-affected area.

There is an existing all-weather roads up to the project site, a 4.0 Km Road is required to be maintained and upgraded for project construction from Powerhouse to the headworks site.

Transmission Line

The electricity generated from the powerhouse will be transmitted to the national grid through a suitable transmission line system. The output power will be evacuated 33kV 3-phase transmission line of approximately 7.5 Km up to NEA Substation, Burtibang, Baglung.

4 INSTALLED CAPACITY AND ENERGY CALCULATION

4.1 Energy Estimation

The proposed Bhuji Khola Hydroelectric Project is classified as a run-of-river (ROR) project that relies on the monthly discharge at the intake site. The hydrological analysis is conducted in compliance with National Electricity Authority (NEA) guidelines, with a minimum of 15% of the total energy produced consisting of dry energy, the energy generated during the dry months. The design discharge is adopted in this study at the 40th percentile of the mean monthly flow, which is set as the design discharge (Qd) of the project, 5.42 m³/s, balancing the balance between dry and wet energy.

Furthermore, following environmental guidelines and regulations, there is a requirement for a minimum downstream release at the intake site. This minimum downstream release is set at 10% of the minimum river flow at the intake site, which is 0.15 m³/s.

Using the designed monthly flow data and available head, the net head is calculated by analyzing the component losses during flow, including structural loss and frictional loss in the tunnel and penstock.

Based on the head available and designed discharge, appropriate turbine units are selected for power generation. The turbine units will be designed to operate continuously between minimum and maximum output and within the head range without exceeding the permissible stresses, vibrations, bearing temperatures, and undue cavitations.

Based on the headworks site and the adopted hydrology, the head loss with electromechanical components and energy calculations in this study are presented in the tables below.

Installed Capacity = $9.81 \times Q_d \times \text{net head} \times \text{eff (turbine} \times \text{generator} \times \text{transformer)}$

Energy = $Q \times t \times (1 - \text{outages}\%)$, Outages assumed are – 4 % for dry & 5% for wet energy

Net head is calculated as Gross head – Total head loss = 170.33 m.

Table 4-1: Energy Calculation Parameter

Weir Crest Elevation	1460.0	masl		Electromechanical Efficiency	
Desander Pond Water Elevation	1460.0	masl		Equipment	Efficiency
Tail Water Elevation	1280.0	masl		Turbine	92%
Gross Head	180.00	m		Generator	97%
Turbine Discharge	5.42	m ³ /s	Q40.0 th	Transformer	99%
Outages Dry Season	4 % Dry & 5 % Wet			Overall Efficiency	88.30%

Month	No. of days in the Month	Flow in the River (m3/s)	Environmental Release (m3/s)	Available Flow of Power Generation (m3/s)	Turbine Discharge	Gross Head	Head Loss	Net Head	Power (MW)	Estimated Energy (with deducting Outage) (KWh)	
										Dry	Wet
January	31	2.46	0.15	2.30	2.30	180.0	4.40	175.60	3,490.94	2,467,396.11	
February	28	2.01	0.15	1.85	1.85	180.0	3.33	176.67	2,825.14	1,803,570.75	
March	31	1.67	0.15	1.52	1.52	180.0	2.60	177.40	2,319.79	1,639,626.39	
April	30	1.52	0.15	1.37	1.37	180.0	2.29	177.71	2,096.69	717,067.43	717,067.43
May	31	1.71	0.15	1.55	1.55	180.0	2.68	177.32	2,376.32		1,679,583.62
June	30	5.83	0.15	5.68	5.42	180.0	9.67	170.33	7,961.07		5,445,374.80
July	31	23.81	0.15	23.66	5.42	180.0	9.67	170.33	7,961.07		5,626,887.29
August	31	33.41	0.15	33.26	5.42	180.0	9.67	170.33	7,961.07		5,626,887.29
September	30	23.90	0.15	23.75	5.42	180.0	9.67	170.33	7,961.07		5,445,374.80
October	31	9.61	0.15	9.45	5.42	180.0	9.67	170.33	7,961.07		5,626,887.29
November	30	4.79	0.15	4.64	4.64	180.0	7.69	172.31	6,899.43		4,719,212.22
December	31	3.13	0.15	2.98	2.98	180.0	5.15	174.85	4,491.69	1,587,363.76	1,587,363.76
									Total	8,215,024.43	32,490,623.69

Hence, the project has a capacity of **7.96 MW**. The estimated generation of dry energy is 8.22 GWh (20.19% of total energy production), while the wet energy generation is projected at 32.49 GWh. Consequently, the total annual energy production is calculated to be 40.71 GWh. The Plant factor is 0.584.

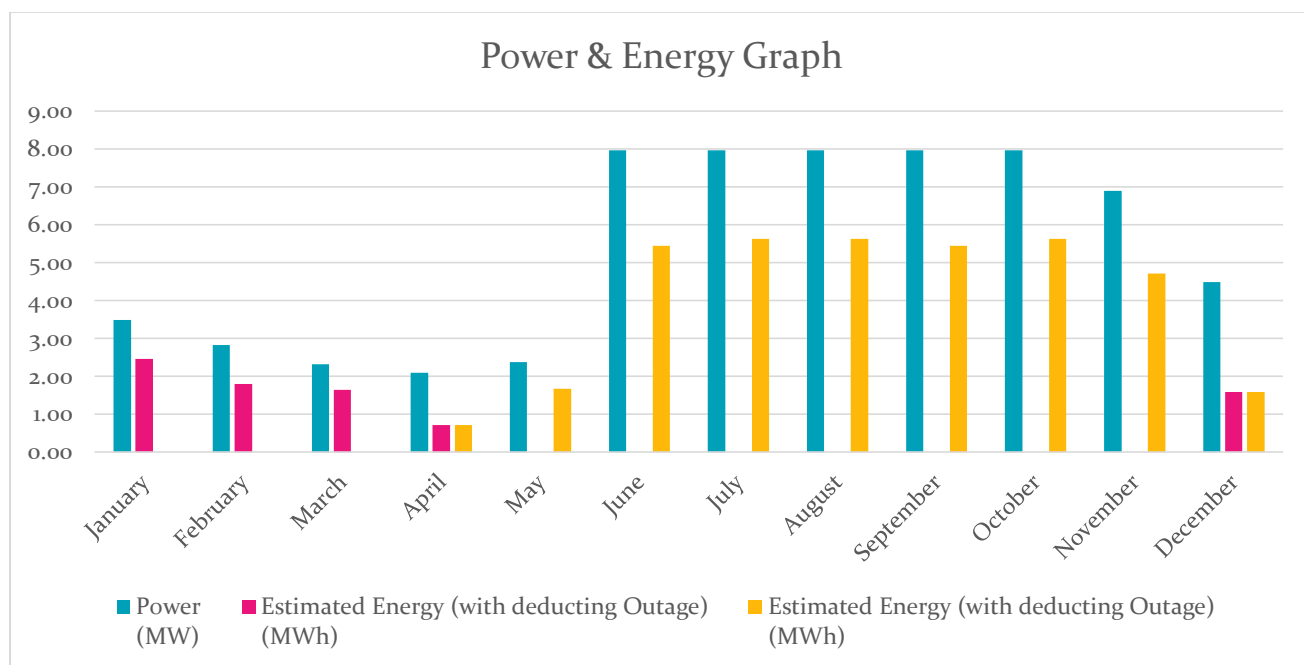


Figure 4-1: Power and Energy Chart for Proposed HEP

5 GEOLOGICAL STUDY OF THE PROJECT

5.1 Introduction

Geological and engineering geological study of the project area is one of the basic requirements for the development of hydropower projects. Such study provides information on the surface and sub-surface condition of rock and soil and their engineering properties.

The proposed Bhuji Khola Hydroelectric Project is located on the upper ridge of the Baglung district of eastern Nepal. The general desk geology report is prepared by using different available sources and maps in the project area.

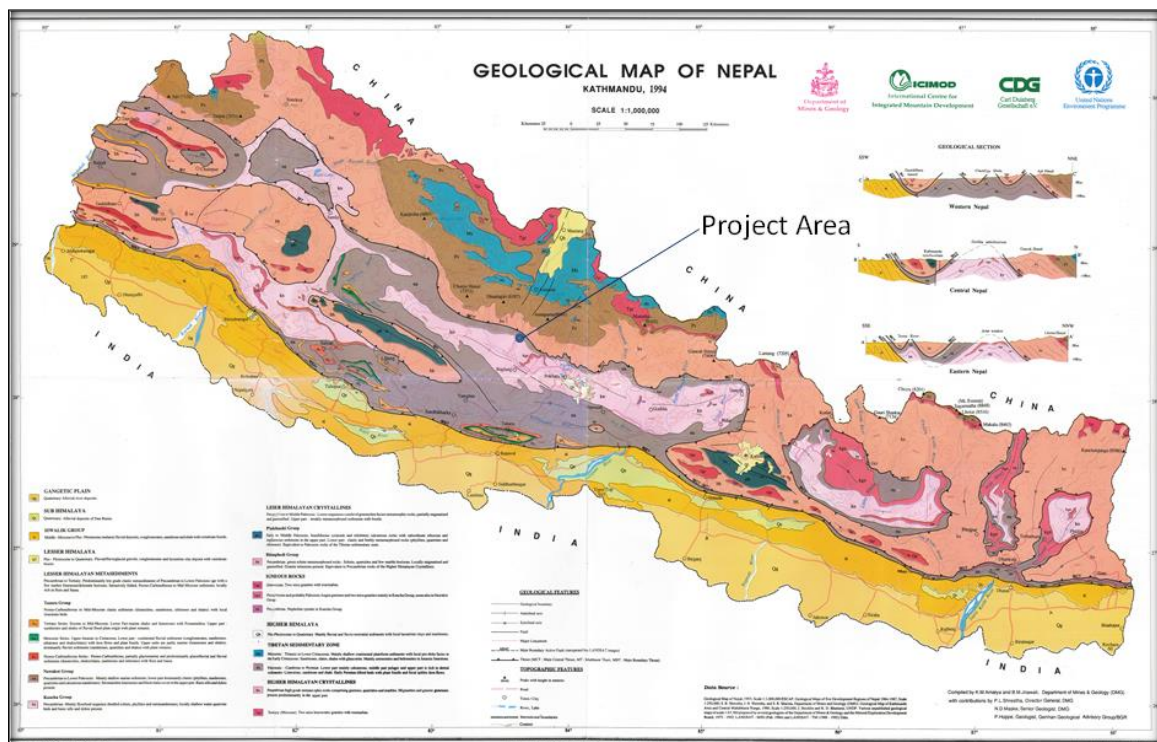


Figure 5-1: Geological Map of Nepal

5.2 Regional Geology

The proposed project area is located in the zone of Lesser Himalaya. The Lesser Himalaya is separated to the north by Main Central Thrust (MCT) from Higher Himalaya and to the south by Main Boundary Thrust (MBT) from Siwalik Group. The Siwalik Group is separated from the Terai Plain by Main Frontal Thrust (MFT). The rocks in the project area are such as Limestone, calcareous quartzite, purple and green shales etc. These rocks have been correlated with the syangja formation of Lakharpata Subgroup of Western Nepal.

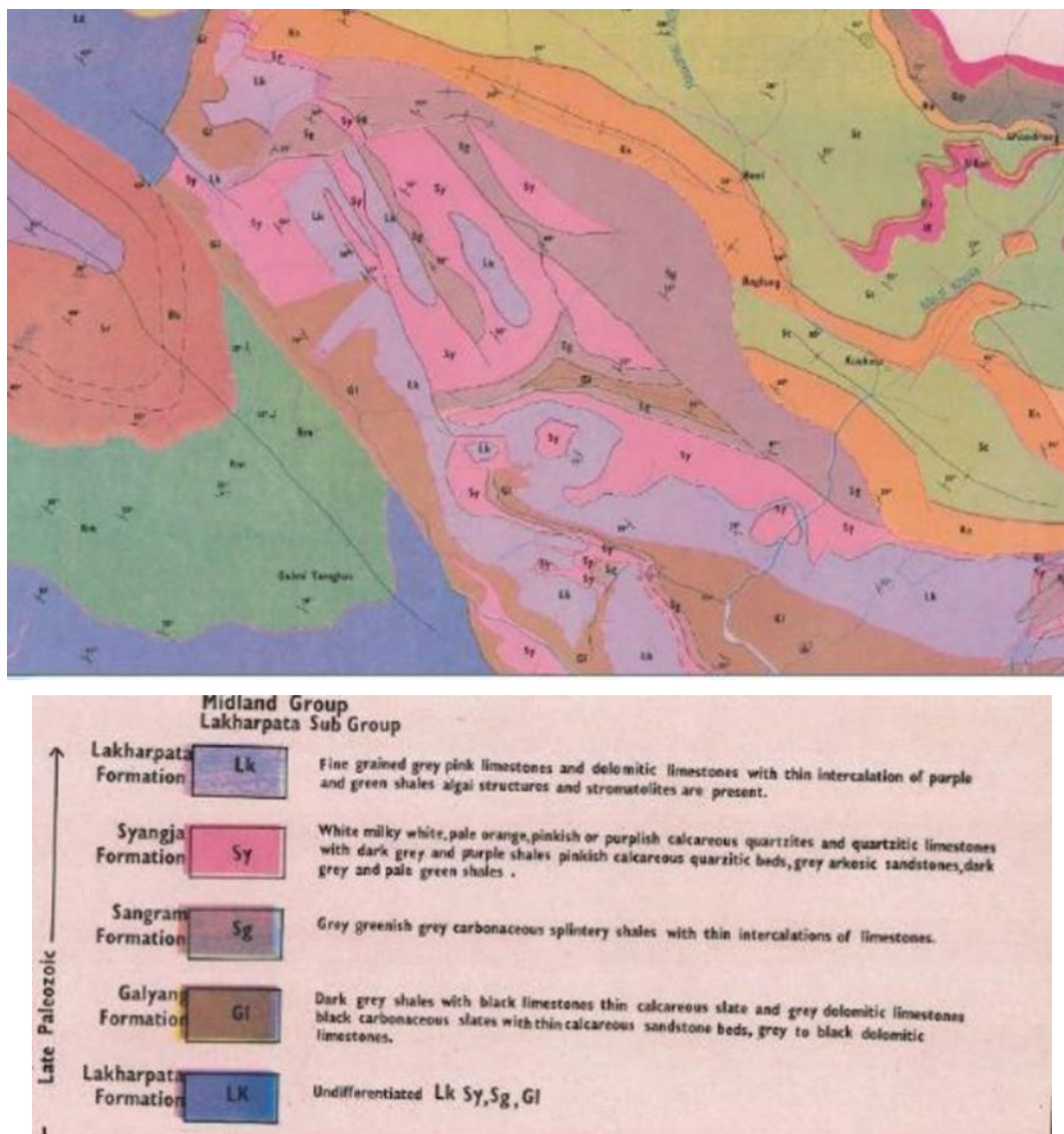


Figure 5-2: Location of Project Area in Regional Geological Map (DMG)

Detailed geological descriptions of the specific structural locations of the different project components will be studied in a detailed study.

5.3 Geo Hazards and Project Risks

The engineering geological aspects in this study of the project site comprised the overall slope stability condition of the project site and observation of the rock and soil mass quality, especially for surface excavation conditions. The overall slope stability condition of the project site is considered fair.

The Bhuji Khola is a normal and medium gradient river in the Lesser Himalaya. No evidences of the debris torrent and mass land sliding were found in and around the project area in this stage of study. The main geo-hazard patterns in the area are extensive surface erosion, slope failures, debris flows and small-scale landslides. The causes of the different patterns of the hazards are land use patterns, steep slope and groundwater action, unplanned development activities such as village road cutting and river bank cutting, respectively.

The altitude variation from the hill tops to the river valley in the area is very low. The average natural slope of the hills is below 35 degrees. The river gradient is also gentle. Consequently, the geo-hazard level of the area is considered to be minimum.

The main observed hazardous processes in the area are:

- Erosional scars downstream of the desander
- Landslides and slope failure at d/s of the pipe alignment.
- Small slope failure by road cutting in the pipe alignment

During the project design, those areas need to be considered safely for the safety of civil structures. The general slope trend of the hill terrain around the project area is gentle. No major slope instability problems will occur during the field survey and mapping.

5.4 Construction Materials

Construction material like gravel will be found on the headworks site of the project area. Coarse aggregates can be obtained by sorting of gravels and crushing of pebbles and cobbles. If the volume of coarse aggregates cannot be sufficient from gravel, pebble and cobble, selective hard boulders should be crushed to fulfill the required volume. Fine aggregate can be extracted from Bhuji Khola and the downstream site of the headworks and power house area.

6 TERMS OF REFERENCE

6.1 Objective of the Feasibility Study

The principal objective of the present study is to carry out a feasibility study including a detailed survey, hydrological study, sediment study, geological study, geophysical investigations, geotechnical investigations, project layout, optimization studies, project component designs, prepare drawings, cost estimates and financial analysis of Bhuji Khola Hydroelectric Project.

The outcome of the study can be used as a basis for securing funding for the project from banks. The company will manage the resources, and frame out the schedule and all associated activities for the execution.

6.2 Scope of the Feasibility Study

To meet the objectives of the study, the effort during the study period will be streamlined covering the scope of services. The study will follow the recommendations of the various international practices and standards on the design of hydropower projects as applicable. The sequence of activities for investigations, surveys, analyses, layout designs, option evaluations, cost estimates, etc. are planned to ensure smooth progress through well-organized link of related activities. The services to be provided by the study team shall be included and not be limited to the following activities:

- Review of available information and Desk Study Reports, Previous Studies
- Establish the Gauge station & complete the hydrological study with sediment study
- Complete geological study with necessary geophysical and geotechnical investigation works
- Conduct feasibility level design and prepare necessary drawings;
- Prepare a cost estimate for the project
- Conduct financial and sensitivity analyses
- Prepare construction plan and project implementation schedule;

6.3 Stages of Work

6.3.1 Field Work

The study team during this stage will perform the following tasks:

- Field visit by the team of experts and engineers
- Site identification/selection
- Detail topographical survey
- Detail geological mapping
- Geophysical and geotechnical investigations
- Hydrological and sediment measurements
- Environment and Social Survey

6.3.2 Map Preparation and Data Analysis

This stage of work shall proceed after the completion of all field works. The study team will process and analyze all the data related to detailed engineering surveys, geological investigations, and hydrological studies. The following tasks will be performed.

- Topographic map preparation
- Geological map preparation and study
- Hydrological analysis and sediment studies

6.3.3 Project Layout Studies

The study team will finalize the layout of the project and project components based on the review of previous works and fieldwork results. The layout plan will include the following:

- Layout of all civil structures such as Dam, sluiceways, intake, settling basin, headrace tunnel, surge tank, penstock, powerhouse, tailrace, and other details.
- Layout of electromechanical equipment inside powerhouse, transmission line layout to connect to the national grid.
- Layout of access to the powerhouse site, and headwork area from the existing road head.

6.3.4 Optimization Study

The optimization study encompasses both the optimization of the project in terms of installed capacity and energy generation in conjunction with the economic as well as environmental and

social aspects. The principal objective of the optimization study is to find out a project configuration, which generates maximum annual energy at the possible lowest cost.

The study team will use several methods and tools to optimize the size of the project components based on working out several options for various structures and alignments of the hydropower project. Optimization procedures for finding the major structures and parameters of the hydropower component will be conducted such that the total costs of construction, operation and maintenance, and mitigation of adverse environmental impacts will be minimized while providing for adequate factors of safety against hydrological, and geological risks.

All of the above activities will include the layout and design of the following components, which are: diversion weir, intake, and water conveyance system; headrace and penstock pipe, fore-bay/surge, powerhouse, mechanical and electrical equipment, power evacuation system, access track, and river flow diversion during construction.

6.3.5 Hydraulic Design of the Project Components

After finding the optimum discharge for the project, the team will prepare the hydraulic design of the project components based on that discharge. This will result in the sizing of the components for the minimum headloss in the system.

6.3.6 Preparation of the Drawings

The drawings of the civil structures will be prepared to the level of feasibility study. The drawings will include the project layout, project component locations and sizing, geological maps, topographical maps, construction material sites, spoil tip areas, location of the campsites, etc.

6.3.7 Preparation of the Bill of Quantities and Rate Analysis

Based on the drawings, a bill of quantities for the project components will be prepared. Similarly, a rate analysis for construction works will be prepared.

6.3.8 Cost Estimate

The cost of the project will be estimated based on the rate analysis and bill of quantities. The cost estimate will include costs for civil works, hydro-mechanical works, electro-mechanical works, transmission line works, infrastructure works, contingencies, and other direct costs.

6.3.9 Financial Evaluation

Financial evaluation and sensitivity analyses will be performed to find the financial viability of the project.

6.4 Environmental Study

As per the Environmental Protection Rules, 1997, the project of this level is legally binding to carry out the IEE and EIA study. EIA is required to study for a hydropower project of capacity above 25 MW and conservation and heritage area.

The study will investigate all the possibilities of direct and indirect effects of the project on the physical, biological, social, and economic environment during and after implementation of the project and recommend appropriate mitigation measures for the direct and indirect effects of the project on the environment. The study will also deal with the land acquisition, compensation, and rehabilitation plan.

A separate volume on the Environmental study report of the Project will be published as part of the Feasibility study.

6.5 Report Preparation

The team shall prepare all the drawings, design cost estimation documents, and prepare a draft final report. The reports shall be consistent with the national requirement of a detailed feasibility study for a ROR type of hydropower project and shall be in a manner acceptable for initiating discussions between the deciding authorities and financial agencies. The team shall be responsible for incorporating all the comments and suggestions made by the concerned authorities and final reports shall be prepared and submitted to them.

6.6 Expected Human Resources

The following professionals will be required to undertake the study. Additional professionals may be involved as deemed necessary.

- Hydropower Experts/Engineers
- Hydraulic Engineers
- Geotechnical Engineers
- Engineering Geologists
- Economists
- Environmental Experts
- Electromechanical Engineers
- Hydro-Mechanical Engineer

- Civil Engineers
- Drafts Person
- Surveyors
- Sociologists

7 BREAK DOWN THE COST OF THE FEASIBILITY STUDY

The breakdown of the cost for the feasibility study and Environmental Impact Assessment of the project is presented in the below table.

Table 7-1: Breakdown of cost for the Feasibility Study and EIA

S.N.	Description of Job	Amount	Remarks
1	Review of Previous Studies and Additional Data Collection	80,000.00	
2	Detail survey of project area	850,000.00	
3	Hydrological analysis	180,000.00	
4	Field visit by the experts	180,000.00	
5	Geological study	200,000.00	
6	Environmental Study	1,200,000.00	
7	Civil component layout, design and alternative optimization	1,170,000.00	
8	Project component layout at site	300,000.00	
9	Structural analysis of all Component	160,000.00	
10	Costing and financial analysis	100,000.00	
11	Electromechanical component design	250,000.00	
12	Hydro mechanical component design	300,000.00	
13	Working drawing preparation	180,000.00	
14	Report Preparation	80,000.00	
	Sub-Total	5,230,000.00	
15	Consultant overhead (5% of cost proposed)	261,500.00	
	Total	5,491,500.00	
	VAT @13%	713,895.00	
	Grand Total	6,205,395.00	

At the time of the feasibility study, the cost may be changed by negotiation with a consultant, and the environmental and technical study team may be a different team.

8 WORK SCHEDULE FOR FEASIBILITY STUDY

S.N	PARTICULARS/MONTHS	1	2	3	4	5	6	7	8	9	10	11	12	14	15	16	17	18	19	20	21	22
1	Maps/Reports Collection and Review of Desk Study	→	→	→																		
2	Field visits by experts		→																			
3	Detail Survey of the Project site		→	→	→																	
4	Topographical Map Preparation				→	→																
5	Civil Structure Design/ Layout and Designs				→	→	→	→	→	→	→	→	→									
6	Geological Map Preparation				→	→	→	→														
7	Drafting											→	→	→	→							
8	Quantity Estimation and Costing of Civil Components										→	→	→	→	→							
9	Electromechanical Design, Estimating and Costing												→	→	→	→	→					
10	Economic Analysis of the Project and Optimization																	→	→			
11	Report Preparation																		→	→		
12	Compilation of the Reports and Drawings																		→	→		
13	Submission of Progress Report						○			○												
14	Submission of Draft Report											○					○			○		
15	Submission of Final Report																	○				○